

BBScore Validation Running tiers 1 through 3

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Tier 1: Environment & Dependencies

Python PASS Python 3.11.14

Dependencies PASS torch PASS numpy PASS scipy PASS sklearn PASS h5py
PASS transformers PASS timm PASS PIL PASS cv2 PASS tqdm PASS boto3
PASS rsatoolbox

BBScore Registries cuML not installed, falling back to scikit-learn. cuML can
speed up linear probes ~50× on GPU; install via: <https://docs.rapids.ai/api/cuml/stable/>
[Warning] Failed to import model module: robust_models No module named
'pkg_resources' PASS 179 models, 165 benchmarks, 22 metrics

Hardware PASS CPU: 10 cores / 10 threads PASS RAM: 34 GB PASS GPU:
Apple MPS (Metal Performance Shaders) PASS Disk: 155 GB free

Configuration PASS SCIKIT_LEARN_DATA = /Users/cml/bbscore_data
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Tier 2: Model Loading & Inference

Vision Model (resnet18) ResNet18 ResNet18 PASS Forward pass on cpu (0.7s)

Vision Model (resnet50) PASS Forward pass on cpu (0.4s)

Language Model (gpt2_small) PASS Forward pass + feature extraction on cpu
(dim=768, 1.6s)

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Tier 3: End-to-End Benchmarks

First run downloads datasets from the cloud. This may take several minutes
depending on your connection.

Vision Benchmark (resnet18 + NSDV1 + RSA) Warning: AWS_CONFIG_FILE
environment variable not set. Authenticated S3 downloads may fail. ResNet18
ResNet18 Extracting features... Running metrics for Separate Layer:
_orig_mod.resnet.encoder.stages.3 Warning: AWS_CONFIG_FILE environ-
ment variable not set. Authenticated S3 downloads may fail. Size of area V1: 6999
Downloading https://storage.googleapis.com/bbscore_datasets/nsd/subj02_nativesurface_nsdgeneral.pkl
to /Users/cml/bbscore_data/NSDAssembly/subj02_nativesurface_nsdgeneral.pkl
using HTTP... Size of area V1: 6417 Downloading https://storage.googleapis.com/bbscore_datasets/nsd/subj05_nativesurface_nsdgeneral.pkl
to /Users/cml/bbscore_data/NSDAssembly/subj05_nativesurface_nsdgeneral.pkl
using HTTP... Size of area V1: 5585 Downloading https://storage.googleapis.com/bbscore_datasets/nsd/subj07_nativesurface_nsdgeneral.pkl
to /Users/cml/bbscore_data/NSDAssembly/subj07_nativesurface_nsdgeneral.pkl
using HTTP... Size of area V1: 5493 PASS ResNet-18 x NSDV1 RSA: spear-
man=0.1562 (1005s)

Language Benchmark (gpt2_small + LeBel2023 + Ridge) Warning:
AWS_CONFIG_FILE environment variable not set. Authenticated

S3 downloads may fail. DEBUG: Found 84 TextGrid files. Extracting features... Running metrics for Separate Layer: transformer.h.11
 Warning: AWS_CONFIG_FILE environment variable not set. Authenticated S3 downloads may fail. Found 0 files for UTS01, expected 84.
 Redownloading... Downloading fMRI data for UTS01... Downloading s3://openneuro.org/ds003020/derivative/preprocessed_data/UTS01/ to /Users/cml/bbscore_data/LeBel2023Assembly/ds003020/derivative/preprocessed_data
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DEBUG: Found 84 HF5 files for UTS01. DEBUG: Stacking 84 stories for UTS01
DEBUG: Final fmri_data shape: (84, 81126) Using Standard K-Fold with 10
splits. PASS GPT-2 Small x LeBel2023 Ridge: median_ceiled_pearson=0.0000
(3584s)

Validation Summary

PASS Tier 1: Environment & Dependencies (8s) PASS Tier 2: Model Loading
& Inference (3s) PASS Tier 3: End-to-End Benchmarks (4588s)

Your machine is ready for BBScore experiments.